

## 1. PROJECT OVERVIEW

- **Project Title:** FitBuddy - AI Workout Generator

## 2. BRAINSTORMING OVERVIEW

The primary objective of the brainstorming session was to design a scalable, web-based health-tech solution that completely automates personalized fitness planning without relying on static templates.

## 3. EVALUATION OF PROPOSED APPROACHES

- **Approach 1: Static Video Library.** A web portal categorized into "Beginner," "Intermediate," and "Advanced" video playlists. (Discarded: Too generic, fails to account for specific user age or weight combinations).
- **Approach 2: Rule-Based Logic Engine.** A Python script with thousands of IF-ELSE conditions that assigns pre-written workouts based on user BMI. (Discarded: Highly rigid, incredibly difficult to maintain, and lacks natural language flexibility).
- **Approach 3: Generative AI Orchestration.** An application that captures user metrics via an HTML form and uses Google Gemini to write a dynamic, completely custom 7-day fitness and nutrition schedule on the fly. (Selected: Highly scalable, deeply personalized, and provides a modern user experience).

**4. FEATURE PRIORITIZATION (CRITICALITY MATRIX)** To ensure a successful MVP (Minimum Viable Product) for the Naan Mudhalvan submission, features were prioritized as follows:

- **Must-Have Features (Core Application):**
  - A responsive frontend form (HTML/Jinja2) to capture Name, Age, Weight, Goal, and Intensity.
  - FastAPI backend routing to process user data efficiently.
  - Integration with Google Gemini API (gemini-3.5-flash-lite) to generate the personalized 7-day workout plan.

- SQLAlchemy & SQLite database integration to store user profiles and the AI-generated plans persistently.
- **Should-Have Features (User Experience):**
  - An Admin Dashboard UI to view a tabular list of all registered users and their corresponding original plans.
  - Exception handling (timeout parameters) to manage API latency without crashing the server.
- **Future Scope (Out of Scope for MVP):**
  - Wearable tech integration (e.g., syncing live heart rate data).
  - Video generation for exercise demonstrations.